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Unit 2: Eigenvalues, Eigenvectors, and Diagonalization of Matrices

Contents

This document presents notes for Unit 2 of the course NMCI102 covering the following topics: Eigenvalues and eigenvectors, Caley–Hamilton theorem, algebraic and geometric multiplicity, (orthogonal) diagonalization, and quadratic forms. We shall always work with real matrices in the notes. Everything can also be studied for complex matrices with appropriate natural modifications.

1 Eigenvalues and Eigenvectors

An $n \times n$ real matrix A can be thought of as the linear mapping that takes any arbitrary vector $x \in \mathbb{R}^n/\mathbb{C}^n$ to the vector Ax. In some cases, the new output vector Ax is simply a scalar multiple of a *non-zero* input vector x, that is, there exists a scalar λ such that $Ax = \lambda x$. This case is so important that we make the following definition.

Definition 1. *Let A be an $n \times n$ matrix. A scalar λ is said to be an* eigenvalue *of A if there exists a nonzero vector $v \in \mathbb{R}^n/\mathbb{C}^n$ such that $Av = \lambda v$, where the vector v is said to be an* eigenvector *of A corresponding to λ.*

Hence, an eigenvector v of A is simply scaled by a scalar λ under multiplication by A. We emphasize that eigenvectors are by definition non-zero vectors.

Example: Determine if the given vectors v and u are eigenvectors of A. If yes, find the eigenvalue of A associated with the eigenvector.

$$A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}, \quad v = \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}, \quad u = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}.$$

Solution: Compute

$$Av = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix} \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -6 \\ 0 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} = 2v.$$

Thus, v is an eigenvector of A with eigenvalue $\lambda = 2$. Hence, $Av = 2v$ and thus v is an eigenvector of A with corresponding eigenvalue $\lambda = 2$. On the other hand,

$$Au = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \\ 4 \end{bmatrix}.$$

There is no scalar λ such that

$$\begin{bmatrix} 0 \\ 6 \\ 4 \end{bmatrix} = \lambda \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}.$$

Therefore, u is not an eigenvector of A.

1.1 The characteristic polynomial of a matrix

Recall that a number λ is an eigenvalue of $A \in \mathbb{R}^{n\times n}$ if there exists a non-zero vector v such that

$$Av = \lambda v,$$

or equivalently, if $v \in$ Null$(\lambda I - A)$. In other words, λ is an eigenvalue of A if and only if the subspace Null$(\lambda I - A)$ contains a vector other than the zero vector. Here I is the identity matrix of size $n \times n$. We know that any $n \times n$ matrix M has a non-trivial null space if and only if M is non-invertible, which is also equivalent to $\det(M) = 0$. Hence, λ is an eigenvalue of A if and only if λ satisfies $\det(\lambda I - A) = 0$.

Let's compute the expression $\det(\lambda I - A)$ for a generic 2×2 matrix:

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - a_{11} & -a_{12} \\ -a_{21} & \lambda - a_{22} \end{vmatrix} = (\lambda - a_{11})(\lambda - a_{22}) - a_{12}a_{21} = \lambda^2 - (a_{11} + a_{22})\lambda + a_{11}a_{22} - a_{12}a_{21}.$$

Thus, if A is 2×2 then

$$\det(\lambda I - A) = \lambda^2 - (a_{11} + a_{22})\lambda + a_{11}a_{22} - a_{12}a_{21}$$

is a polynomial in the variable λ of degree $n = 2$. This motivates the following definition.

Definition 2. *Let A be an $n \times n$ matrix. The polynomial*

$$p(\lambda) = \det(\lambda I - A)$$

is called the characteristic polynomial of A.

In summary, to find the eigenvalues of A we must find the roots of the characteristic polynomial:

$$p(\lambda) = \det(\lambda I - A).$$

The following theorem asserts that what we observed for the case $n = 2$ is indeed true for all n.

Theorem 3. *The characteristic polynomial $p(\lambda) = \det(\lambda I - A)$ of an $n \times n$ matrix A is an nth degree monic polynomial.*

Proof. Recall that for the case $n = 2$ we computed that

$$\det(\lambda I - A) = \lambda^2 - (a_{11} + a_{22})\lambda + a_{11}a_{22} - a_{12}a_{21}.$$

Therefore, the claim holds for $n = 2$. By induction, suppose that the claims hold for $n \geq 2$. Let A be an $(n+1) \times (n+1)$ matrix. Denote by A_{1i} the $n \times n$ submatrix of A obtained by deleting the first row and ith column. By expanding $\det(\lambda I - A)$ along the first row:

$$\det(\lambda I - A) = (\lambda - a_{11}) \det(\lambda I - A_{11}) + \sum_{k=2}^{n+1} (-1)^k a_{1k} \det(\lambda I - A_{1k}).$$

By the induction hypothesis, each of $\det(\lambda I - A_{1k})$ is an nth degree polynomial. Moreover, $(\lambda - a_{11}) \det(\lambda I - A_{11})$ is an $(n+1)$th degree monic polynomial. Therefore, $\det(\lambda I - A)$ is an $(n+1)$th degree monic polynial. The statement of the theorem thus holds by mathematical induction. $\square$

Example: Find the eigenvalues of

$$A = \begin{pmatrix} -4 & -6 & -7 \\ 3 & 5 & 3 \\ 0 & 0 & 3 \end{pmatrix}.$$

Solution. Compute

$$\lambda I - A = \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} - \begin{pmatrix} -4 & -6 & -7 \\ 3 & 5 & 3 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} \lambda + 4 & 6 & 7 \\ -3 & \lambda - 5 & -3 \\ 0 & 0 & \lambda - 3 \end{pmatrix}.$$

Then

$$\begin{aligned}\det(\lambda I - A) &= (\lambda+4)\begin{vmatrix}\lambda-5 & -3 \\ 0 & \lambda-3\end{vmatrix} - 6\begin{vmatrix}-3 & -3 \\ 0 & \lambda-3\end{vmatrix} + 7\begin{vmatrix}-3 & \lambda-5 \\ 0 & 0\end{vmatrix} \\ &= (\lambda+4)(\lambda-5)(\lambda-3) + 18(\lambda-3) \\ &= [(\lambda+4)(\lambda-5)+18](\lambda-3) \\ &= \left[\lambda^2-\lambda-20+18\right](\lambda-3) \\ &= \left[\lambda^2-\lambda-2\right](\lambda-3) \\ &= (\lambda+1)(\lambda-2)(\lambda-3).\end{aligned}$$

Therefore, the eigenvalues of A are $\lambda_1 = 2, \lambda_2 = 3, \lambda_3 = -1$.

Definition 4. *Let A be an $n \times n$ matrix and λ be an eigenvalue of A. Then the eigenspace of A corresponding to λ is defined by*

$$E_\lambda := \mathrm{Null}(A - \lambda I) = \{u \in \mathbb{R}^n : Au = \lambda u\}. \tag{1}$$

Example: Find the eigenvalues of A and a basis for each eigenspace.

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 4 & 2 & 2 \\ -2 & 0 & 1 \end{bmatrix}$$

Does $\mathbb{R}^3$ have a basis consisting of only eigenvectors of A?
Solution. The characteristic polynomial of A is

$$p(\lambda) = \det(\lambda I - A) = \lambda^3 - 5\lambda^2 + 8\lambda - 4 = (\lambda-1)(\lambda-2)^2,$$

and therefore the eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 2$. Notice that although $p(\lambda)$ is a polynomial of degree $n = 3$, it has only two distinct roots and hence A has only two distinct eigenvalues. The eigenvalue $\lambda_2 = 2$ is said to be repeated and $\lambda_1 = 1$ is said to be a simple eigenvalue.

For $\lambda_1 = 1$, one finds that the eigenspace $\text{Null}(\lambda_1 I - A)$ is spanned by

$$v_1 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

and thus v_1 is an eigenvector of A with eigenvalue $\lambda_1 = 1$.

Now consider $\lambda_2 = 2$:

$$2I - A = \begin{bmatrix} 0 & 0 & 0 \\ -4 & 0 & -2 \\ 2 & 0 & 1 \end{bmatrix}.$$

Row reducing $2I - A$, one obtains

$$2I - A \sim \begin{bmatrix} 2 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Therefore, $\text{rank}(2I - A) = 1$, and thus by the rank-nullity theorem it follows that $\text{Null}(2I - A)$ is a 2-dimensional eigenspace. Performing back substitution, one finds the following basis for the λ_2-eigenspace:

$$\{v_2, v_3\} = \left\{ \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Therefore, the eigenvectors

$$\{v_1, v_2, v_3\} = \left\{ \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

form a basis for $\mathbb{R}^3$. Hence, for the repeated eigenvalue $\lambda_2 = 2$, we were able to find two linearly independent eigenvectors.

1.2 Cayley–Hamilton theorem

Theorem 5 (Cayley–Hamilton theorem)**.** *Every square matrix satisfies its own characteristic equation.*

The Cayley–Hamilton theorem can be used to compute

- the positive integral power of an $n \times n$ matrix, and
- the inverse of a non-singular square matrix.

Example: Verify Cayley–Hamilton theorem for the following matrix:

$$A = \begin{bmatrix} -2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}.$$

Also, compute A^4 and A^{-1}. **Solution:** The characteristic polynomial of A is given by $\det(\lambda I - A) = \lambda^3 - 2\lambda^2 - 8\lambda + 9$, which can be obtained by simple algebraic calculations.

The Cayley–Hamilton theorem states that the following equation holds:

$$A^3 - 2A^2 - 8A + 9I = 0. \tag{1}$$

Verification:

$$A^2 = A \times A = \begin{bmatrix} -2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} -2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & -2 & 1 \\ -1 & 6 & -6 \\ 1 & -5 & 7 \end{bmatrix}.$$

$$A^3 = A \times A^2 = \begin{bmatrix} -2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 7 & -2 & 1 \\ -1 & 6 & -6 \\ 1 & -5 & 7 \end{bmatrix} = \begin{bmatrix} -11 & -12 & 18 \\ -10 & 19 & -20 \\ 10 & -18 & 21 \end{bmatrix}.$$

We have,

$$A^3-2A^2-8A+9I = \begin{bmatrix} -11 & -12 & 18 \\ -10 & 19 & -20 \\ 10 & -18 & 21 \end{bmatrix} -2 \begin{bmatrix} 7 & -2 & 1 \\ -1 & 6 & -6 \\ 1 & -5 & 7 \end{bmatrix} -8 \begin{bmatrix} -2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} +9 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

To find A^4: From (1),

$$A^3 = 2A^2 + 8A - 9I. \tag{2}$$

Multiply A on both sides, we get

$$A^4 = 2A^3 + 8A^2 - 9A.$$

This gives,

$$A^4 = 2 \begin{bmatrix} -11 & -12 & 18 \\ -10 & 19 & -20 \\ 10 & -18 & 21 \end{bmatrix} + 8 \begin{bmatrix} 7 & -2 & 1 \\ -1 & 6 & -6 \\ 1 & -5 & 7 \end{bmatrix} - 9 \begin{bmatrix} -2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 52 & -31 & 26 \\ -19 & 68 & -79 \\ 19 & -67 & 80 \end{bmatrix}.$$

To find A^{-1}: Multiply (1) by A^{-1},

$$A^2 - 2A - 8I + 9A^{-1} = 0$$

$$9A^{-1} = -A^2 + 2A + 8I$$

$$9A^{-1} = - \begin{bmatrix} 7 & -2 & 1 \\ -1 & 6 & -6 \\ 1 & -5 & 7 \end{bmatrix} + 2 \begin{bmatrix} -2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} + 8 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & 0 & 3 \\ -1 & 6 & 4 \\ 1 & 3 & 5 \end{bmatrix}.$$

So,

$$A^{-1} = \frac{1}{9}\begin{bmatrix} -3 & 0 & 3 \\ -1 & 6 & 4 \\ 1 & 3 & 5 \end{bmatrix}.$$

2 Algebraic multiplicity and Geometric multiplicity

Definition 6. *Suppose that $A \in M_{n\times n}$ has characteristic polynomial $p(\lambda)$ that can be factored as*

$$p(\lambda) = (\lambda - \lambda_1)^{k_1}(\lambda - \lambda_2)^{k_2} \cdots (\lambda - \lambda_p)^{k_p},$$

where $\lambda_1, \ldots, \lambda_p$ are distinct eigenvalues of A. The exponent k_i is called the algebraic multiplicity of the eigenvalue λ_i. The dimension Null$(\lambda_i I - A)$ of the eigenspace associated to λ_i is called the geometric multiplicity of λ_i.

For simplicity and whenever it is convenient, we will denote the geometric multiplicity of the eigenvalue λ_i as

$$g_i = \dim(\textbf{Null}(\lambda_i I - A)).$$

Remark 1. *We emphasize that the characteristic polynomial of a real matrix can have non-real roots. However, we will only call the real roots as the eigenvalues of the matrix because we are only focusing on real eigenvalues and real eigenvectors. In principle, the theory extends to the complex field as well, but we shall restrict ourselves to the real field only.*

Example: if a 6×6 matrix A has characteristic polynomial

$$p(\lambda) = \lambda^6 - 4\lambda^5 - 12\lambda^4,$$

then find the eigenvalues of A and their algebraic multiplicities.

Solution. Factoring $p(\lambda)$ we obtain

$$p(\lambda) = \lambda^4(\lambda^2 - 4\lambda - 12) = \lambda^4(\lambda - 6)(\lambda + 2).$$

Therefore, the eigenvalues of A are $\lambda_1 = 0$, $\lambda_2 = 6$, and $\lambda_3 = -2$. Their algebraic multiplicities are $k_1 = 4$, $k_2 = 1$, and $k_3 = 1$, respectively. The eigenvalue $\lambda_1 = 0$ is repeated, while $\lambda_2 = 6$ and $\lambda_3 = -2$ are simple eigenvalues.

In the above example, we had $p(\lambda) = (\lambda - 1)(\lambda - 2)^2$ and thus $\lambda_1 = 1$ has algebraic multiplicity $k_1 = 1$ and $\lambda_2 = 2$ has algebraic multiplicity $k_2 = 2$. For $\lambda_1 = 1$, we found one linearly independent eigenvector, and therefore λ_1 has geometric multiplicity $g_1 = 1$. For $\lambda_2 = 2$, we found two linearly independent eigenvectors, and therefore λ_2 has geometric multiplicity $g_2 = 2$. However, as we will see in the next example, the geometric multiplicity g_i is in general less than the algebraic multiplicity k_i:

$$g_i \le k_i.$$

Example: Find the eigenvalues of A and a basis for each eigenspace:

$$A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

For each eigenvalue of A, find its algebraic and geometric multiplicity. Does $\mathbb{R}^3$ have a basis of eigenvectors of A?

Solution. One computes

$$p(\lambda) = \lambda^3 + 3\lambda^2 - 4 = (\lambda - 1)(\lambda + 2)^2$$

and therefore the eigenvalues of A are $\lambda_1 = 1$ and $\lambda_2 = -2$. The algebraic multiplicity of λ_1 is $k_1 = 1$ and that of λ_2 is $k_2 = 2$.

For $\lambda_1 = 1$, we compute

$$\lambda_1 I - A = \begin{bmatrix} -1 & -4 & -3 \\ 4 & 7 & 3 \\ -3 & -3 & 0 \end{bmatrix}$$

and then one finds that

$$v_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

is a basis for the λ_1-eigenspace. Therefore, the geometric multiplicity of λ_1 is $g_1 = 1$.

For $\lambda_2 = -2$, we compute

$$\lambda_2 I - A = \begin{bmatrix} -4 & -4 & -3 \\ 4 & 4 & 3 \\ -3 & -3 & -3 \end{bmatrix} \sim \begin{bmatrix} 4 & 4 & 3 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Therefore, since $\operatorname{rank}(\lambda_2 I - A) = 2$, the geometric multiplicity of $\lambda_2 = -2$ is $g_2 = 1$, which is less than the algebraic multiplicity $k_2 = 2$.

An eigenvector corresponding to $\lambda_2 = -2$ is

$$v_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

Therefore, for the repeated eigenvalue $\lambda_2 = -2$, we are able to find only one linearly independent eigenvector. Therefore, it is not possible to construct a basis for $\mathbb{R}^3$ consisting of eigenvectors of A.

3 Eigenvalues and Similarity Transformations

Definition 7. *Let A and B be $n \times n$ matrices. We will say that A is similar to B if there exists an invertible matrix P such that*

$$A = PBP^{-1}.$$

If A is similar to B then B is similar to A because from the equation $A = PBP^{-1}$ we can multiply on the left by P^{-1} and on the right by P to obtain that

$$P^{-1}AP = B.$$

Hence, with $Q = P^{-1}$, we have that $B = QAQ^{-1}$ and thus B is similar to A. Hence, if A is similar to B then B is similar to A and therefore we simply say that A and B are similar.

Matrices that are similar are clearly not necessarily equal. However, there is a reason why the word similar is used. Here are a few reasons why.

Theorem 8. *If A and B are similar matrices then the following are true:*

1. *rank*$(A) =$ *rank*(B)
2. $\det(A) = \det(B)$
3. *A and B have the same characteristic polynomial and hence the same eigenvalues*

Proof. We will prove part (c). If A and B are similar then $A = PBP^{-1}$ for some matrix P. Then

$$\det(A - \lambda I) = \det(A - \lambda PP^{-1}) = \det(PBP^{-1} - \lambda PP^{-1}) = \det(P(B - \lambda I)P^{-1})$$

Using properties of determinants:

$$= \det(P)\det(B - \lambda I)\det(P^{-1}) = \det(B - \lambda I).$$

Thus, A and B have the same characteristic polynomial, and hence the same eigenvalues. □

Remark 2. *The converse of Theorem 7 (3) need not be true, i.e., if the characteristic polynomials of A and B are same then A need not be similar to B. For example, let*

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Both the matrices have their characteristic polynomials $p(x) = x^2$, but they are not similar (why?).

Example: Examine whether A is similar to B, where

(a) $A = \begin{bmatrix} 5 & 5 \\ -2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix}$ $\quad (b)$ $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

Solution. The given matrices are similar if there exists an invertible matrix P such that $A = P^{-1}BP$. Let $P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. We shall determine a, b, c and d such that $PA = BP$ and then check whether P is non-singular.

(a). Consider $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 5 & 5 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. This is equivalent to

$$\begin{bmatrix} 5a - 2b & 5a \\ 5c - 2d & 5c \end{bmatrix} = \begin{bmatrix} a + 2c & b + 2d \\ -3a + 4c & -3b + 4d \end{bmatrix}$$

Equating the corresponding elements, we obtain the system of equations

$$\begin{aligned} 4a - 2b - 2c &= 0 \\ 5a - b - 2d &= 0 \\ 3a + c - 2d &= 0 \\ 3b + 5c - 4d &= 0. \end{aligned}$$

A solution to this system of equations is $a = b = c = 1, d = -2$. Therefore, we get $P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$ and $det(P) = -3 \neq 0$. Hence, P is nonsingular and $A = P^{-1}BP$.

(b). Again, consider $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. This is equivalent to

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a + c & b + d \\ c & d \end{bmatrix}$$

Equating the corresponding elements, we get

$$a = a + c, \quad b = b + d \quad \text{or} \quad c = d = 0.$$

Hence, $P = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}$, which is a singular matrix. Since an invertible matrix P does not exist, the matrices A and B are not similar.

Remark 3. *In general, for given matrices A and B, finding the matrix P is a difficult problem. However, it is possible to obtain the matrix P when A or B is a diagonal matrix. Thus, our interest is to find a matrix P such that for a given matrix A, we have $D = P^{-1}AP$ or $PDP^{-1} = A$, where D is a diagonal matrix*

4 Diagonalization

4.1 Eigenvalues of triangular matrices

Before discussing diagonalization, we first consider the eigenvalues of triangular matrices.

Theorem 9. *Let A be a triangular matrix (either upper or lower). Then the eigenvalues of A are its diagonal entries.*

Proof. We will prove the theorem for the case $n = 3$ and A is upper triangular; the general case is similar. Suppose then that A is a 3×3 upper triangular matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

Then

$$\lambda I - A = \begin{bmatrix} \lambda - a_{11} & -a_{12} & -a_{13} \\ 0 & \lambda - a_{11} & -a_{23} \\ 0 & 0 & \lambda - a_{11} \end{bmatrix}$$

and thus the characteristic polynomial of A is

$$p(\lambda) = \det(\lambda I - A) = -(a_{11} - \lambda)(a_{22} - \lambda)(a_{33} - \lambda)$$

and the roots of $p(\lambda)$ are

$$\lambda_1 = a_{11}, \quad \lambda_2 = a_{22}, \quad \lambda_3 = a_{33}.$$

In other words, the eigenvalues of A are simply the diagonal entries of A. $\square$

Example: Consider the following matrix:

$$A = \begin{bmatrix} 6 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 & 0 \\ -1 & 0 & 0 & -4 & 0 \\ 8 & -2 & 3 & 0 & 7 \end{bmatrix}$$

1. Find the characteristic polynomial and the eigenvalues of A.
2. Find the geometric and algebraic multiplicity of each eigenvalue of A.

We now introduce a very special type of a triangular matrix, namely, a diagonal matrix.

Definition 10. *A square matrix whose off-diagonal entries are all zero is called a diagonal matrix.*

For example, here is a 3×3 diagonal matrix:

$$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & -8 \end{bmatrix}$$

and here is a 5×5 diagonal matrix:

$$D = \begin{bmatrix} 6 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -7 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}$$

A diagonal matrix is clearly also a triangular matrix, and therefore the eigenvalues of a diagonal matrix D are simply the diagonal entries of D. Moreover, the powers of a diagonal matrix are easy to compute. For example, if

$$D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

then

$$D^2 = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} \lambda_1^2 & 0 \\ 0 & \lambda_2^2 \end{bmatrix}$$

and similarly for any integer $k = 1, 2, 3, \ldots$, we have that

$$D^k = \begin{bmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{bmatrix}.$$

4.2 Diagonalizable matrices

Definition 11. *A matrix A is called diagonalizable if it is similar to a diagonal matrix D. In other words, if there exists an invertible matrix P such that*

$$A = PDP^{-1}.$$

How do we determine when a given matrix A is diagonalizable? Let us first determine what conditions must be met for a matrix A to be diagonalizable. Suppose that A is diagonalizable. Then by Definition, there exists an invertible matrix

$$P = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix}$$

and a diagonal matrix

$$D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

such that $A = PDP^{-1}$. Multiplying on the right both sides of the equation $A = PDP^{-1}$ by the matrix P, we obtain

$$AP = PD.$$

Now

$$AP = \begin{bmatrix} Av_1 & Av_2 & \cdots & Av_n \end{bmatrix}$$

while on the other hand

$$PD = \begin{bmatrix} \lambda_1 v_1 & \lambda_2 v_2 & \cdots & \lambda_n v_n \end{bmatrix}.$$

Therefore, since it holds that $AP = PD$, we have

$$\begin{bmatrix} Av_1 & Av_2 & \cdots & Av_n \end{bmatrix} = \begin{bmatrix} \lambda_1 v_1 & \lambda_2 v_2 & \cdots & \lambda_n v_n \end{bmatrix}.$$

Comparing columns, we must have that

$$Av_i = \lambda_i v_i.$$

Thus, the columns $v_1, v_2, \ldots, v_n$ of P are eigenvectors of A and form a basis for $\mathbb{R}^n$ because P is invertible. In conclusion, if A is diagonalizable, then $\mathbb{R}^n$ has a basis consisting of eigenvectors of A.

Suppose instead that $\{v_1, v_2, \ldots, v_n\}$ is a basis of $\mathbb{R}^n$ consisting of eigenvectors of A. Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the eigenvalues of A associated with $v_1, v_2, \ldots, v_n$, respectively, and set

$$P = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix}.$$

Then P is invertible because $\{v_1, v_2, \ldots, v_n\}$ are linearly independent. Let

$$D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}.$$

Now, since $Av_i = \lambda_i v_i$ we have that

$$AP = A\begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} = \begin{bmatrix} Av_1 & Av_2 & \cdots & Av_n \end{bmatrix} = \begin{bmatrix} \lambda_1 v_1 & \lambda_2 v_2 & \cdots & \lambda_n v_n \end{bmatrix}.$$

Therefore,

$$AP = \begin{bmatrix} \lambda_1 v_1 & \lambda_2 v_2 & \cdots & \lambda_n v_n \end{bmatrix}.$$

On the other hand,

$$PD = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} = \begin{bmatrix} \lambda_1 v_1 & \lambda_2 v_2 & \cdots & \lambda_n v_n \end{bmatrix}.$$

Therefore, $AP = PD$, and since P is invertible we have that

$$A = PDP^{-1}.$$

Thus, if $\mathbb{R}^n$ has a basis consisting of eigenvectors of A, then A is diagonalizable. We have therefore proved the following theorem.

Theorem 12. *A matrix A is diagonalizable if and only if there is a basis $\{v_1, v_2, \ldots, v_n\}$ of $\mathbb{R}^n$ consisting of eigenvectors of A.*

Here, the punchline is that the problem of diagonalization of a matrix A is equivalent to finding a basis of $\mathbb{R}^n$ consisting of eigenvectors of A. We will see in some of the examples below that it is not always possible to diagonalize a matrix.

4.3 Sufficient conditions for diagonalizability

We first consider the simplest case when we conclude that a given matrix is diagonalizable, namely, the case when all eigenvalues are distinct.

Theorem 13. *Suppose that $A \in \mathbb{R}^{n\times n}$ has n distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then A is diagonalizable.*

Proof. Each eigenvalue λ_i produces an eigenvector v_i. The set of eigenvectors $\{v_1, v_2, \ldots, v_n\}$ are linearly independent because they correspond to distinct eigenvalues. Therefore, $\{v_1, v_2, \ldots, v_n\}$ is a basis of $\mathbb{R}^n$ consisting of eigenvectors of A, and then by Theorem 5 we conclude that A is diagonalizable. □

If A does not have distinct eigenvalues, can A still be diagonalizable? The following theorem completely answers this question.

Theorem 14. *A matrix A is diagonalizable if and only if the algebraic and geometric multiplicities of each eigenvalue are equal.*

Proof. Let A be an $n\times n$ matrix and let $\lambda_1, \lambda_2, \ldots, \lambda_p$ denote the distinct eigenvalues of A. Suppose that $k_1, k_2, \ldots, k_p$ are the algebraic multiplicities and $g_1, g_2, \ldots, g_p$ are the geometric multiplicities of the eigenvalues, respectively. Suppose that the algebraic and geometric multiplicities of each eigenvalue are equal, that is, suppose that $g_i = k_i$ for each $i = 1, 2, \ldots, p$. Since $k_1 + k_2 + \cdots + k_p = n$, then because $g_i = k_i$ we must also have that $g_1 + g_2 + \cdots + g_p = n$. Therefore, there exists n linearly independent eigenvectors of A and consequently A is diagonalizable.

On the other hand, suppose that A is diagonalizable. Since the geometric multiplicity is at most the algebraic multiplicity, the only way that $g_1 + g_2 + \cdots + g_p = n$ is if $g_i = k_i$, i.e., that the geometric and algebraic multiplicities are equal. $\square$

Example: Determine if A is diagonalizable. If yes, find a matrix P that diagonalizes A.

$$A = \begin{bmatrix} -4 & -6 & -7 \\ 3 & 5 & 3 \\ 0 & 0 & 3 \end{bmatrix}$$

Solution. The characteristic polynomial of A is

$$p(\lambda) = \det(A - \lambda I) = (\lambda - 2)(\lambda - 3)(\lambda + 1)$$

and therefore $\lambda_1 = 2, \lambda_2 = 3$, and $\lambda_3 = -1$ are the eigenvalues of A. Since A has $n = 3$ distinct eigenvalues, so A is diagonalizable. Eigenvectors v_1, v_2, v_3 corresponding to $\lambda_1, \lambda_2, \lambda_3$ are found to be

$$v_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

Therefore, a matrix that diagonalizes A is

$$P = \begin{bmatrix} -1 & -2 & -2 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

You can verify that

$$P \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} P^{-1} = A$$

Example: Determine if A is diagonalizable. If yes, find a matrix P that diagonalizes A.

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 4 & 2 & 2 \\ -2 & 0 & 1 \end{bmatrix}$$

Solution. The characteristic polynomial of A is

$$p(\lambda) = \det(A - \lambda I) = (\lambda - 1)(\lambda - 2)^2$$

and therefore $\lambda_1 = 1, \lambda_2 = 2$. An eigenvector corresponding to $\lambda_1 = 1$ is

$$v_1 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

One finds that $g_2 = \dim(\textbf{Null}(A - \lambda_2 I)) = 2$, and two linearly independent eigenvectors for λ_2 are

$$\{v_2, v_3\} = \left\{ \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Therefore, A is diagonalizable, and a matrix that diagonalizes A is

$$P = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ -2 & 0 & 1 \\ 1 & 2 & 0 \end{bmatrix}$$

You can verify that

$$P \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} P^{-1} = A$$

4.4 Orthogonal diagonalization

Definition 15. *Let* $v = (v_1, v_2, \ldots, v_n)^T$ *and* $u = (u_1, u_2, \ldots, u_n)^T$ *be two vectors in* $\mathbb{R}^n$*. Then we say* v *is* **Orthogonal** *to* u *if* $v^T.u = \sum_{i=1}^{n} v_i u_i = 0$.

Recall that an $n \times n$ matrix A is diagonalizable if and only if it has n linearly independent eigenvectors. Moreover, the matrix P with these eigenvectors as columns is a diagonalizing matrix for A, that is,

$$P^{-1}AP \text{ is diagonal.}$$

As we have seen, the really nice bases of $\mathbb{R}^n$ are the orthogonal ones, so a natural question is: which $n \times n$ matrices have an orthogonal basis of eigenvectors? These turn out to be precisely the symmetric matrices, and this is the main result of this section.

Before proceeding, recall that an orthogonal set of vectors is called *orthonormal* if $\|v\| = 1$ for each vector v in the set, and that any orthogonal set $\{v_1, v_2, \ldots, v_k\}$ can be "normalized," that is, converted into an orthonormal set:

$$\left\{ \frac{v_1}{\|v_1\|}, \frac{v_2}{\|v_2\|}, \ldots, \frac{v_k}{\|v_k\|} \right\}.$$

In particular, if a matrix A has n orthogonal eigenvectors, they can (by normalizing) be taken to be orthonormal. The corresponding diagonalizing matrix P has orthonormal columns, and such matrices are very easy to invert.

Theorem 16. *The following conditions are equivalent for an* $n \times n$ *matrix* P.

1. *P is invertible and* $P^{-1} = P^T$.
2. *The rows of* P *are orthonormal.*
3. *The columns of* P *are orthonormal.*

Proof. First, recall that condition (1) is equivalent to $PP^T = I$. Let $x_1, x_2, \ldots, x_n$ denote the rows of P. Then x_j^T is the jth column of P^T, so the (i, j)-entry of PP^T is given by $x_i \cdot x_j$. Thus, $PP^T = I$ means that $x_i \cdot x_j = 0$ if $i \neq j$ and $x_i \cdot x_j = 1$ if $i = j$. Hence, condition (1) is equivalent to (2). The proof of the equivalence of (1) and (3) is similar. $\square$

Definition 17. *An $n \times n$ matrix P is called an **orthogonal matrix** if it satisfies one (and hence all) of the conditions in Theorem 16.*

Example: The rotation matrix

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

is orthogonal for any angle θ.

Example: The matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$$

has orthogonal rows, but the columns are not orthogonal. However, if the rows are normalized, the resulting matrix

$$A' = \begin{bmatrix} \frac{\sqrt{2}}{6} & \frac{\sqrt{1}}{6} & \frac{\sqrt{1}}{6} \\ \frac{\sqrt{-1}}{3} & \frac{\sqrt{1}}{3} & \frac{\sqrt{1}}{3} \\ 0 & \frac{\sqrt{-1}}{2} & \frac{\sqrt{1}}{2} \end{bmatrix}$$

is orthogonal (so the columns are now orthonormal).

Example: If P and Q are orthogonal matrices, then PQ is also orthogonal, as is $P^{-1} = P^T$.

Solution: Since P and Q are invertible, their product PQ is also invertible, and

$$(PQ)^{-1} = Q^{-1}P^{-1} = Q^T P^T = (PQ)^T.$$

Hence, PQ is orthogonal. Similarly, $(P^{-1})^{-1} = P = (P^T)^T = (P^{-1})^T$ shows that P^{-1} is orthogonal.

Definition 18. *An $n \times n$ matrix A is said to be* orthogonally diagonalizable *when an orthogonal matrix P can be found such that $P^{-1}AP = P^T AP$ is diagonal.*

Theorem 19. *The following conditions are equivalent for an $n \times n$ matrix A.*

1. *A has an orthonormal set of n eigenvectors.*
2. *A is orthogonally diagonalizable.*
3. *A is symmetric.*

5 Quadratic Forms

Definition 20. *A quadratic form in n variables is a function $f : \mathbb{R}^n \to \mathbb{R}$ of the form*

$$f(x) = f(x_1, \ldots, x_n) = \sum_{1 \le i \le j \le n} c_{ij} x_i x_j$$

where $x \in \mathbb{R}^n$ and $c_{ij} \in \mathbb{R}$ for $1 \le i \le j \le n$. Alternatively, a quadratic form is a homogeneous polynomial of degree 2 in n variables $x_1, \ldots, x_n$.

Examples: The following are quadratic forms:

1. $f(x_1) = x_1^2$
2. $f(x_1, x_2) = 2x_1^2 + 3x_2^2 - x_1 x_2$
3. $f(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2 - 2x_1x_2 - 2x_1x_3 - 2x_2x_3$
4. $f(x_1, \ldots, x_n) = x_1^2 + x_2^2 + \cdots + x_n^2 = \langle v, v \rangle$, where $v = (x_1, \ldots, x_n)$.

Thus, the Euclidean inner product gives rise to a quadratic form.

If we set $a_{ii} = c_{ii}$ for $i = 1, \ldots, n$ and $a_{ij} = \frac{1}{2} c_{ij}$ for $1 \le i < j \le n$, then

$$f(x) = \sum_{i=1}^{n} a_{ii} x_i^2 + \sum_{1 \le i < j \le n} 2a_{ij} x_i x_j.$$

This can be written in matrix form as

$$f(x) = x^T A x,$$

where A is the symmetric $n \times n$ matrix with (i, j)-th entry equal to a_{ij}. This matrix A is called the matrix of the quadratic form f.

Example: Let

$$f(x_1, x_2) = 2x_1^2 - 3x_2^2 - x_1x_2.$$

Then,

$$f(x_1, x_2) = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2 & -\frac{1}{2} \\ -\frac{1}{2} & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Since a symmetric matrix is involved, we can use Theorem 9, which states that there exists an orthogonal matrix Q such that

$$Q^T AQ = D = \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix},$$

where D is a diagonal matrix, and $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A.

Setting $y = Q^{-1}x = Q^T x$, we obtain

$$x = Qy,$$

so that

$$f(x) = (Qy)^T A(Qy) = y^T Q^T AQy = y^T Dy.$$

If

$$y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix},$$

then

$$y^T Dy = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2,$$

which is a quadratic form in the variables $y_1, \ldots, y_n$ with no cross terms. This process is called the *diagonalization* of the quadratic form f.

Theorem 21. ***(Principal Axes Theorem)*** *Every quadratic form f can be diagonalized. Specifically, if*

$$f(x) = x^T Ax$$

is a quadratic form in

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix},$$

then there exists an orthogonal matrix Q *such that*

$$f(x) = x^T Ax = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2,$$

where

$$\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = Q^T \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix},$$

and $\lambda_1, \dots, \lambda_n$ *are the eigenvalues of* A.

From linear algebra, we know that Q is the matrix whose columns are the unit eigenvectors of A.

Example: Let

$$f(x_1, x_2, x_3) = 2x_1x_2 + 2x_1x_3 + 2x_2x_3.$$

The matrix of f is

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

The eigenvalues of A are $2, -1, -1$ with corresponding unit eigenvectors

$$\begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}, \quad \begin{bmatrix} -2/\sqrt{6} \\ 1/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}.$$

Thus,

$$Q = \begin{bmatrix} 1/\sqrt{3} & -2/\sqrt{6} & 0 \\ 1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{6} & -1/\sqrt{2} \end{bmatrix}.$$

Setting $y = Q^T x$, we get

$$y_1 = \frac{1}{\sqrt{3}}(x_1 + x_2 + x_3), \quad y_2 = \frac{1}{\sqrt{6}}(-2x_1 + x_2 + x_3), \quad y_3 = \frac{1}{\sqrt{2}}(x_2 - x_3).$$

Expressing f in terms of y_1, y_2, y_3, we obtain

$$f(y) = 2y_1^2 - y_2^2 - y_3^2.$$

Definition 22. *A quadratic form $f : \mathbb{R}^n \to \mathbb{R}$ is called* positive definite *if*

$$f(x) > 0 \quad \textit{for all } x \neq 0.$$

An immediate consequence of the Principal Axes Theorem is the following:

Theorem 23. *Let $f(x) = x^T Ax$ be a quadratic form with matrix A. Then f is positive definite if and only if all the eigenvalues of A are positive.*

Proof. By the Principal Axes Theorem, there exists an orthogonal matrix Q such that

$$f(x) = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2,$$

where

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = Q^T x$$

and $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A. If all the λ_i are positive, then $f(x) > 0$ except when $y = 0$. But this happens if and only if $x = 0$ because Q^T is invertible. Therefore, f is positive definite. On the other hand, if one of the eigenvalues $\lambda_i \leq 0$, letting $y = e_i$ and $x = Qy$, we get $f(x) = \lambda_i \leq 0$ and so f is not positive definite. $\square$

We say that a symmetric matrix A is positive definite if the associated quadratic form

$$f(x) = x^T Ax$$

is positive definite. The Principal Axes Theorem has important applications in geometry.